B27-L6 lab: Register, serve, containerise and monitor a model

Platform: Laptop

Work in `labs/serving/` in this repository. You register model versions in MLflow and serve the champion through an alias, write a FastAPI service with a health route and a validated prediction route, package it as a Docker image, and measure drift with PSI. You need Python 3.12, Docker and the lab's dev extras on the laptop. Deploying to Azure is optional and needs a subscription. The running service and its README are the roadmap's evidence for month 18: one deployed model.

1. Run `cd labs/serving`, create a virtual environment, then `pip install -e ".[dev]"`. Run `pytest`: most tests fail with NotImplementedError. Read `README.md` and every file in `src/serve_lab/`.
2. Write `psi` and `drift_status` in `drift.py` (lesson B27-L5). Run `pytest tests/test_drift_and_model.py`.
3. Write `register`, `promote` and `load_alias` in `registry.py` (lesson B27-L1), then `create_app` in `api.py` (lesson B27-L2). Run `pytest`: every test passes.
4. Set `MLFLOW_TRACKING_URI=sqlite:///mlflow.db`. Run `python -m serve_lab train` twice, once with `--c 0.01`, then `promote 1`, `export` and `serve`. Open `http://127.0.0.1:8000/docs` and send a request.
5. Build and run the image (lesson B27-L3): `docker build -t survey-topics:1 .`, then `docker run --rm -p 8000:8000 survey-topics:1`. Call `/health` and `/predict` and read the JSON log line.
6. Run `python -m serve_lab drift` and compare with the README's reference table. Then promote version 2, export, rebuild as `survey-topics:2`, and roll back to version 1 (lesson B27-L6).
7. Optional: deploy with `az containerapp up` (lesson B27-L4), call the public URL, then delete the resource group. Write your README:

the registry steps, the API contract, the image, the drift report and the rollback. Push it to a new GitHub repository.

Notes